

CALIPER UNIFIED GIS & MODELING NETWORK SOLUTIONS

Caliper®



CALIPER CORPORATION

- Leading developer of travel demand modeling software solutions in the U.S.
 - Used in 36 states by over 180 MPOs, more than any other software
 - Used in over 80 countries around the world
- Only U.S. owned and operated developer of travel demand and traffic simulation software products
- GIS-based transportation software solutions
- Also offering custom consulting services and software development



NETWORK MANAGEMENT GOALS

- Up-to-date Statewide model base years
- Up-to-date MPO model base years
- Easy management of alternative future scenarios for MPO and Statewide networks
- Easy coding of future projects
- Consistency with HPMS ARNOLD LRS



A UNIFIED MASTER NETWORK SOLUTION

- A single, master network
 - With accurate, routable line work
 - With support for multiresolution modeling
 - For management of projects and scenarios
 - In a collaborative environment



WHY A UNIFIED, MASTER NETWORK APPROACH?

- Quality assurance and quality control
 - All eyes on the same thing
 - Less duplication of work and opportunities for human error
- Efficiency and Cost Effectiveness
 - Cheaper to develop and maintain a single solution
 - Easier for staff to learn one tool / system than a collection
 - Collaboration between ODOT & MPO staff and elimination of duplicative efforts
- Consistency
 - Built-in consistency of assumptions for project coding, base assumptions, etc.



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ACCURATE, ROUTABLE LINE WORK

- Leverage Existing Data Resources
- Ensure Routability & Topological Consistency
- Leverage Caliper's Existing Tools & Expertise



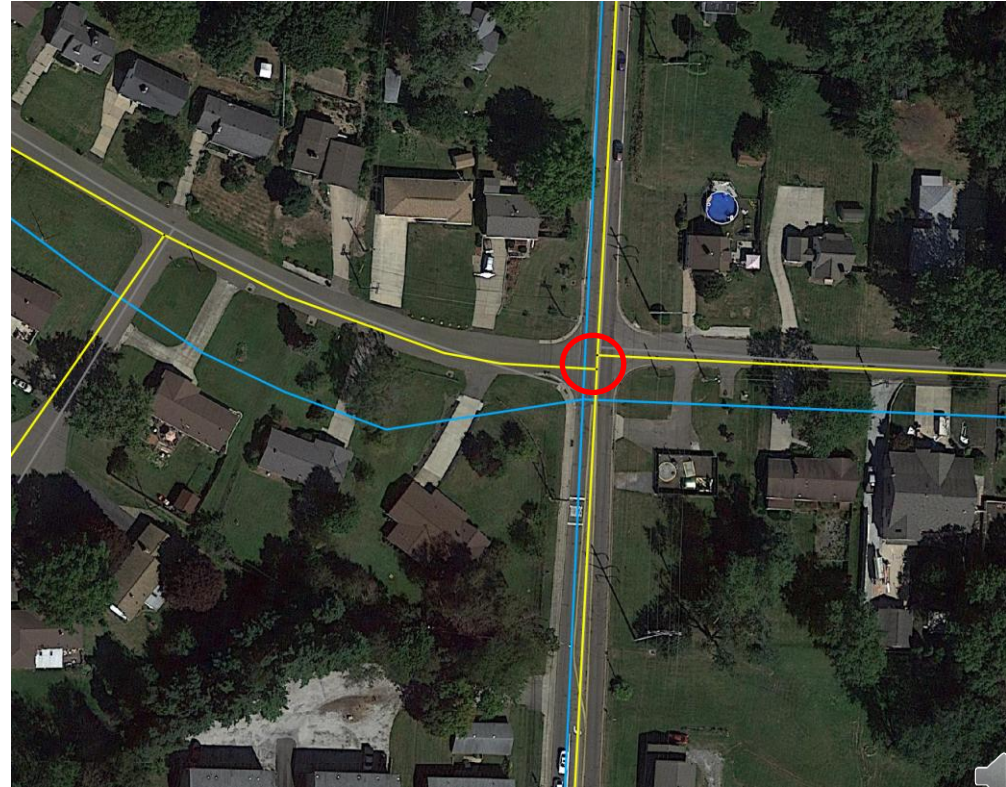
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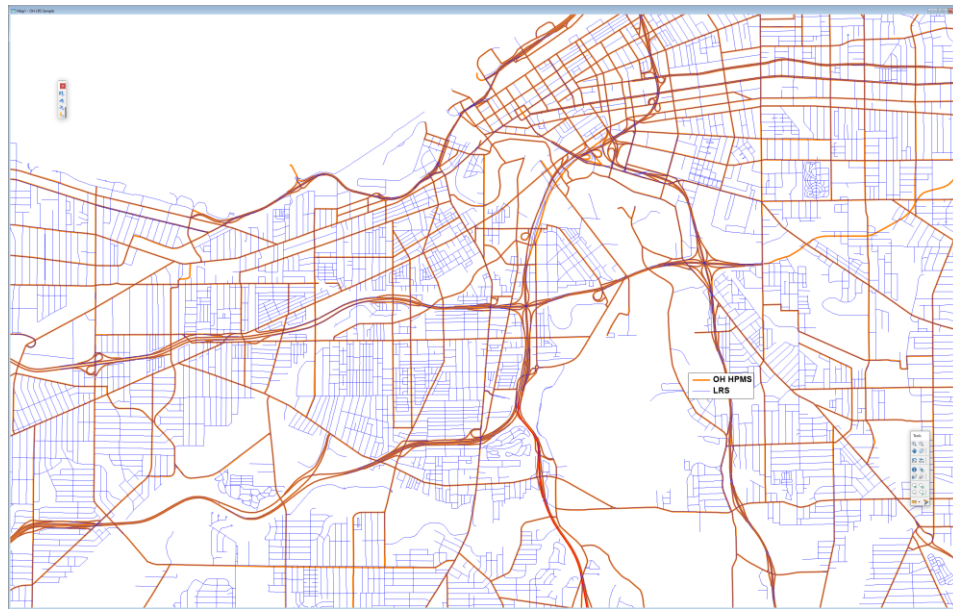
LEVERAGING EXISTING DATA RESOURCES

- Existing model network
 - blue
- ODOT's ARNOLD line work
 - yellow



LEVERAGING EXISTING DATA RESOURCES

- Existing model network
- ODOT's ARNOLD line work
- E911 network
- Tiger lines



LEVERAGING EXISTING DATA RESOURCES

- Existing model network
- ODOT's ARNOLD line work
- E911 network
- Tiger lines
- Caliper networks
 - **National Highway Network** with functional class, FHWA designations, and HPMS data suitable for statewide models
 - **National Rail Network** suitable for flowing waybills and estimating traffic
 - **TransModeler Network** Lane-level road networks for microscopic traffic simulation with interchange and intersection detail, currently all freeways and some major roads



LEVERAGING EXISTING DATA RESOURCES

- Existing model network
- ODOT's ARNOLD line work
- E911 network
- Tiger lines
- Caliper networks
- HERE networks
 - Caliper is a licensed reseller of HERE networks and data which provides a high quality, elegant solution.
 - But many agencies don't like HERE licensing restrictions.



ACCURATE, ROUTABLE LINE WORK

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ROUTABILITY & TOPOLOGICAL CONSISTENCY

- Importance of Nodes in travel modeling
 - For continuity and routability links have to meet at a node
 - For associations between data tables / structures
(e.g., zones & centroids, intersections and configurations / turn penalties)
- TransCAD's GIS enforces link-node topological linkage
 - Lack of native node handling in most other GIS solutions



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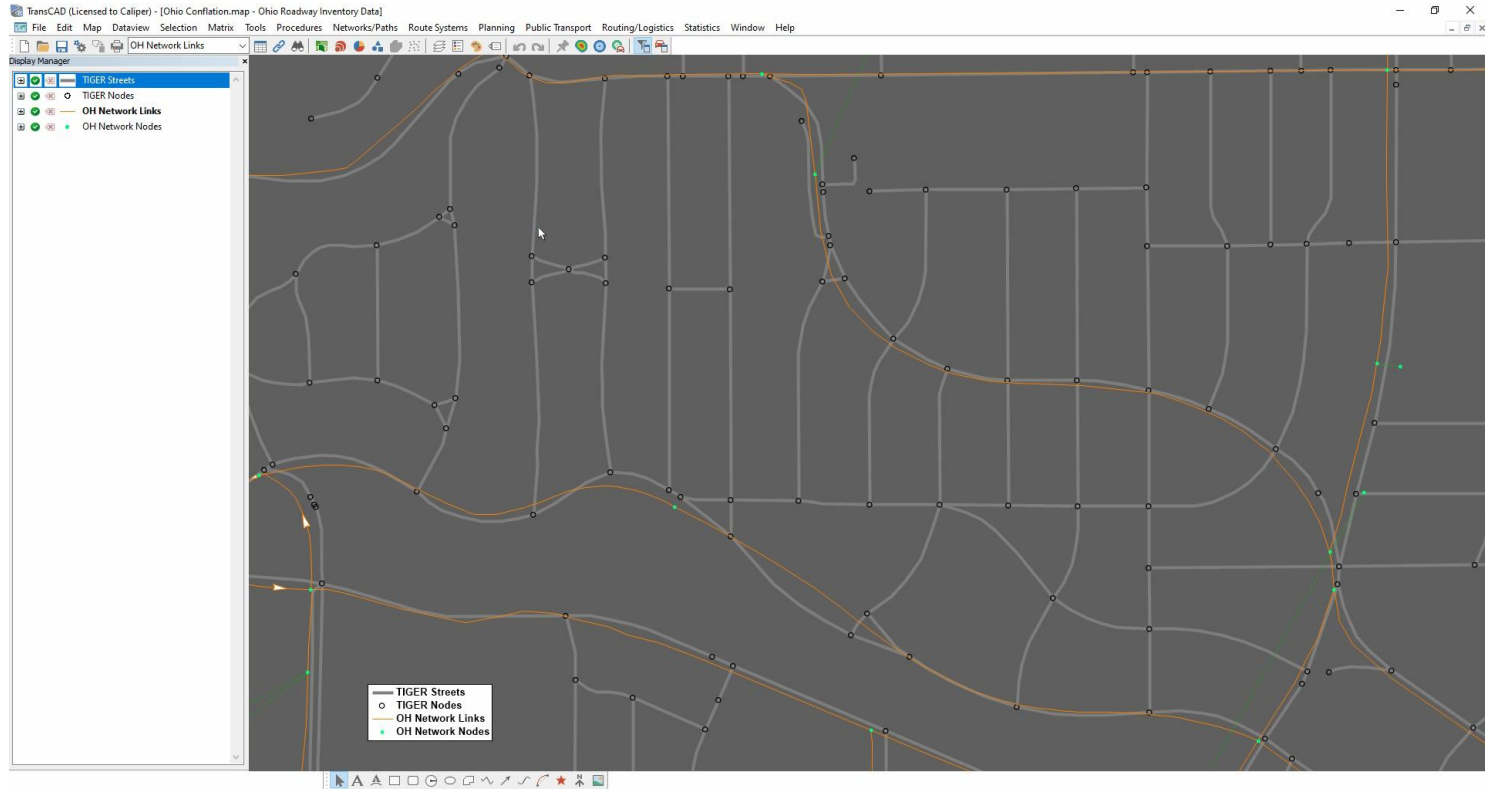


CALIPER'S BUILT-IN TOOLS IN TRANSCAD

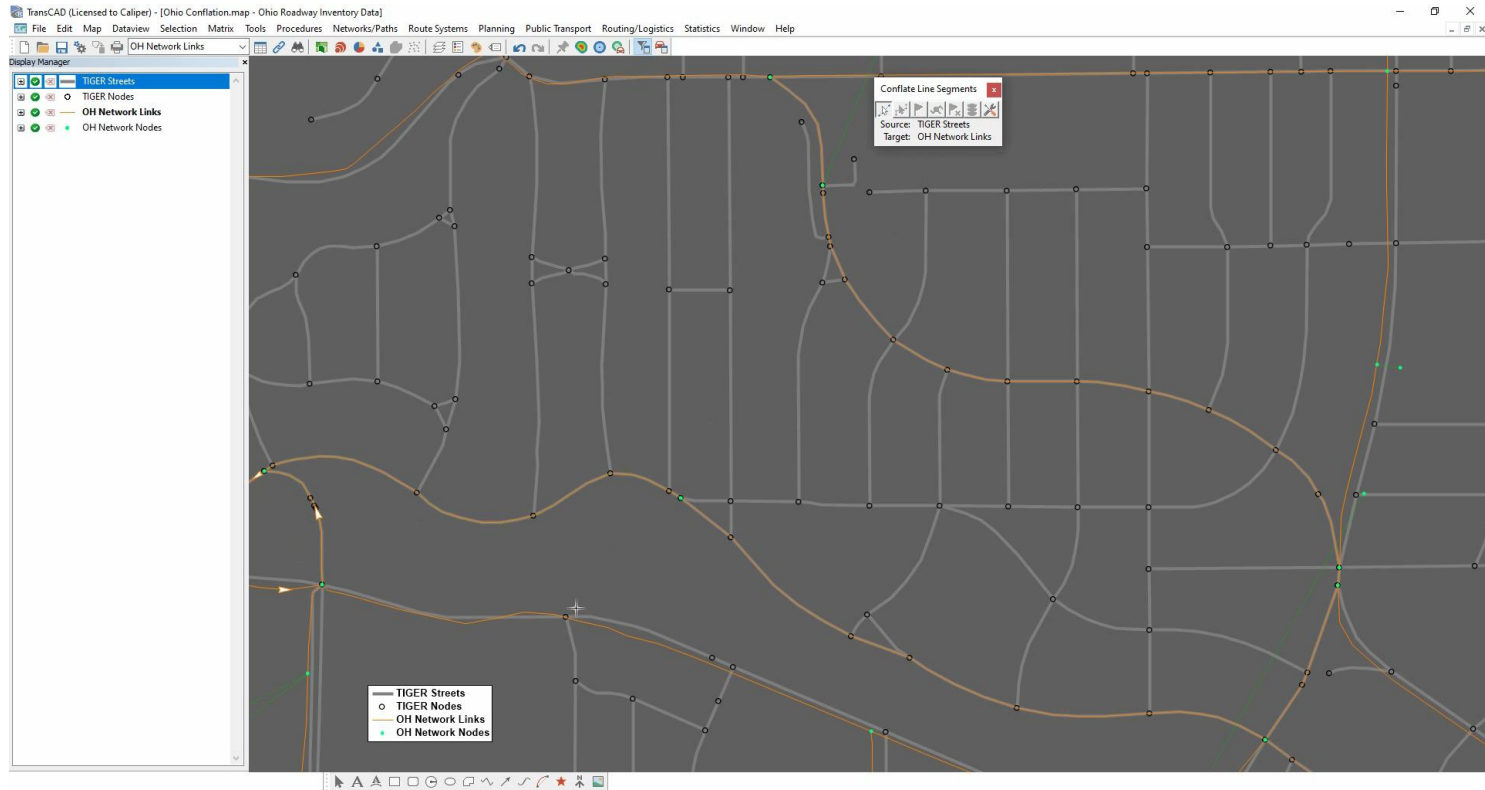
- Convert from lane-level networks to centerline geography
- Add, Delete, & Reshape over Imagery
- Copy & Paste between line layers
- Conflation
- Dualization
- Interchange Creation from Templates
- Linear Referencing System (LRS) data attachment tools



TRANSCAD CONFLATION TOOLS



TRANSCAD CONFLATION TOOLS



CALIPER'S EXPERTISE

- Caliper routinely develops networks
 - FHWA, FRA, etc.
- In-house tools in addition to built-in TransCAD functionality
 - Additional special conflation tools
 - Efficient adjustment and validation to imagery
- Low-cost editing and network correction



TOLEDO MPO NETWORK – BEFORE & AFTER



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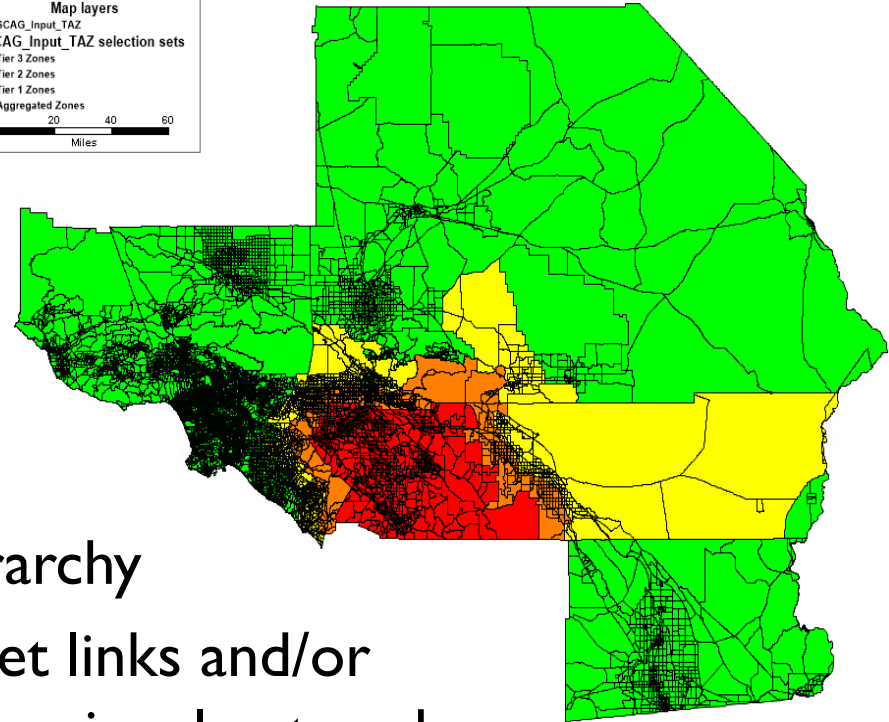
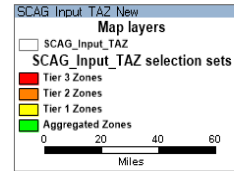
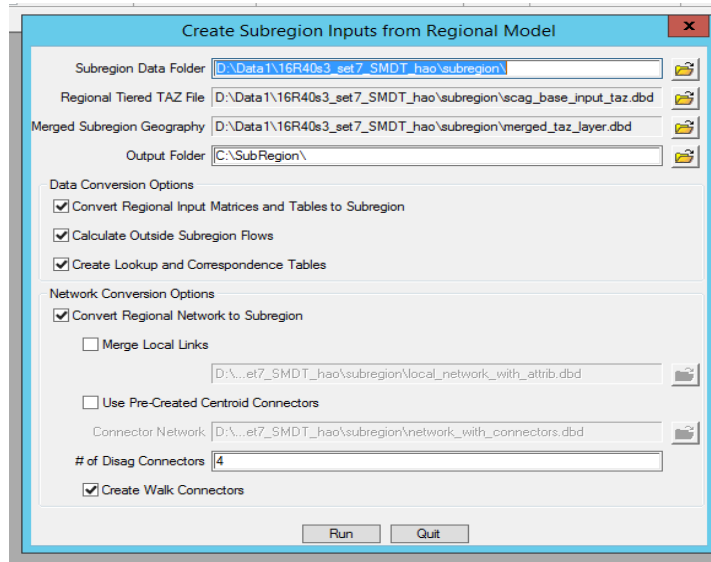


SUPPORT FOR MULTIREOLUTION MODELING

- Ohio modeling networks as subsets of master network
 - Statewide network
 - MPO networks
 - Focus networks
 - Nonmotorized networks
- Automated removal of extraneous nodes with respect for chosen link attributes creates fast networks after export
- Aggregate zones and socioeconomic data on the fly



SCAG SUBAREA MODELING TOOL



- Predefined, nesting zone hierarchy
- Options to import local street links and/or pre-defined connectors into regional network



INTEGRATED ALL-STREETS AND MICROZONES



- Manage all-streets networks and highway assignment networks in a unified framework
- Manage zones and microzones in a unified framework
- Use all-streets for walk-access to transit and/or walk, bike skimming and assignments at microzone level



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MANY MASTER NETWORK SOLUTIONS

- TransCAD permits flexible solutions aided by strong scenario management capabilities
- Several implemented by Caliper and various consultants
- Some handle attributes based on additional fields, better ones based on auxiliary project & scenario files
- Different alternatives for managing geography
- Enabling and Disabling Links is a useful feature
- As ODOT knows, a more comprehensive solution is needed to ensure evolutionary consistency

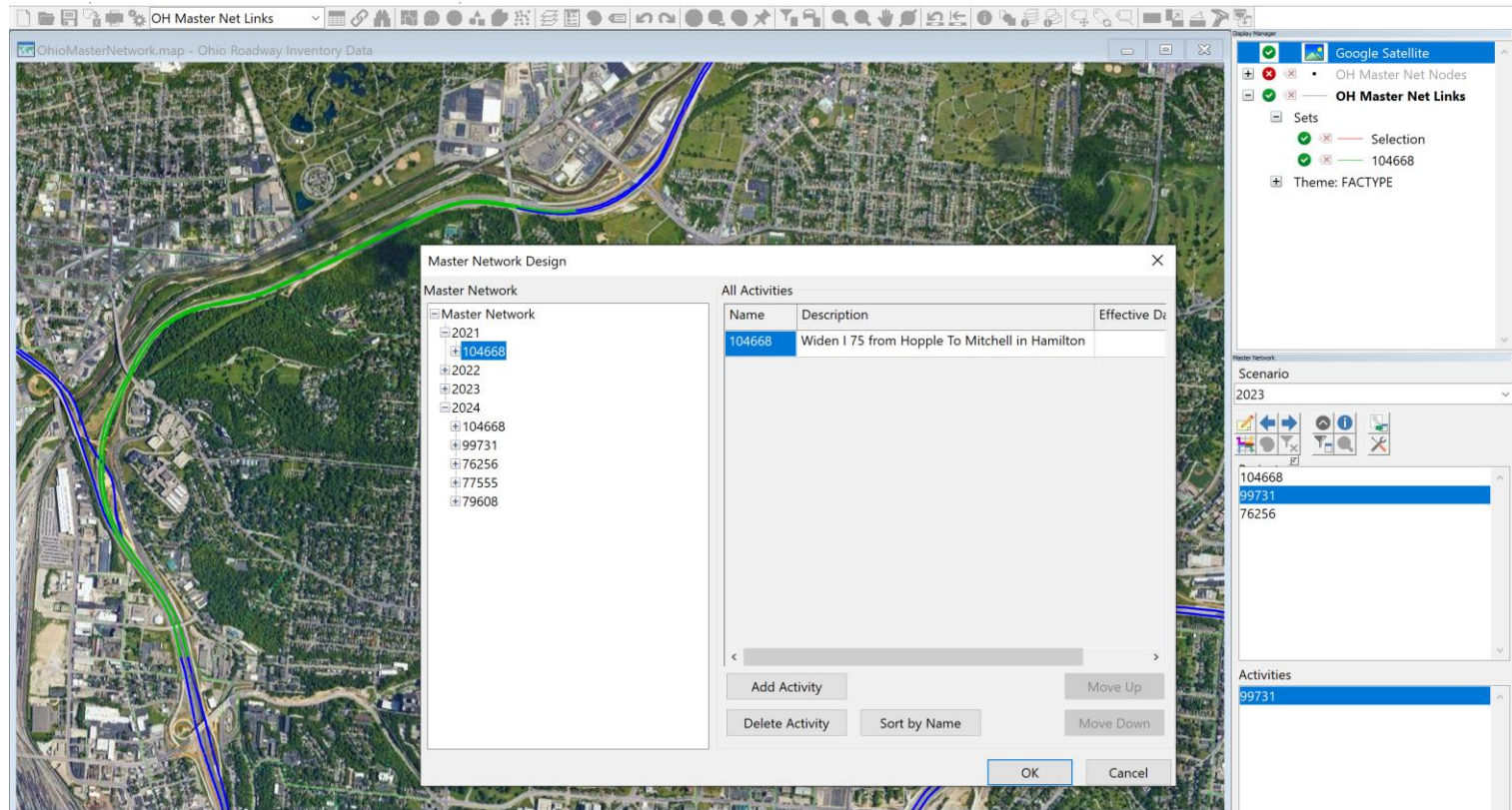


TRANSCAD MASTER NETWORKS

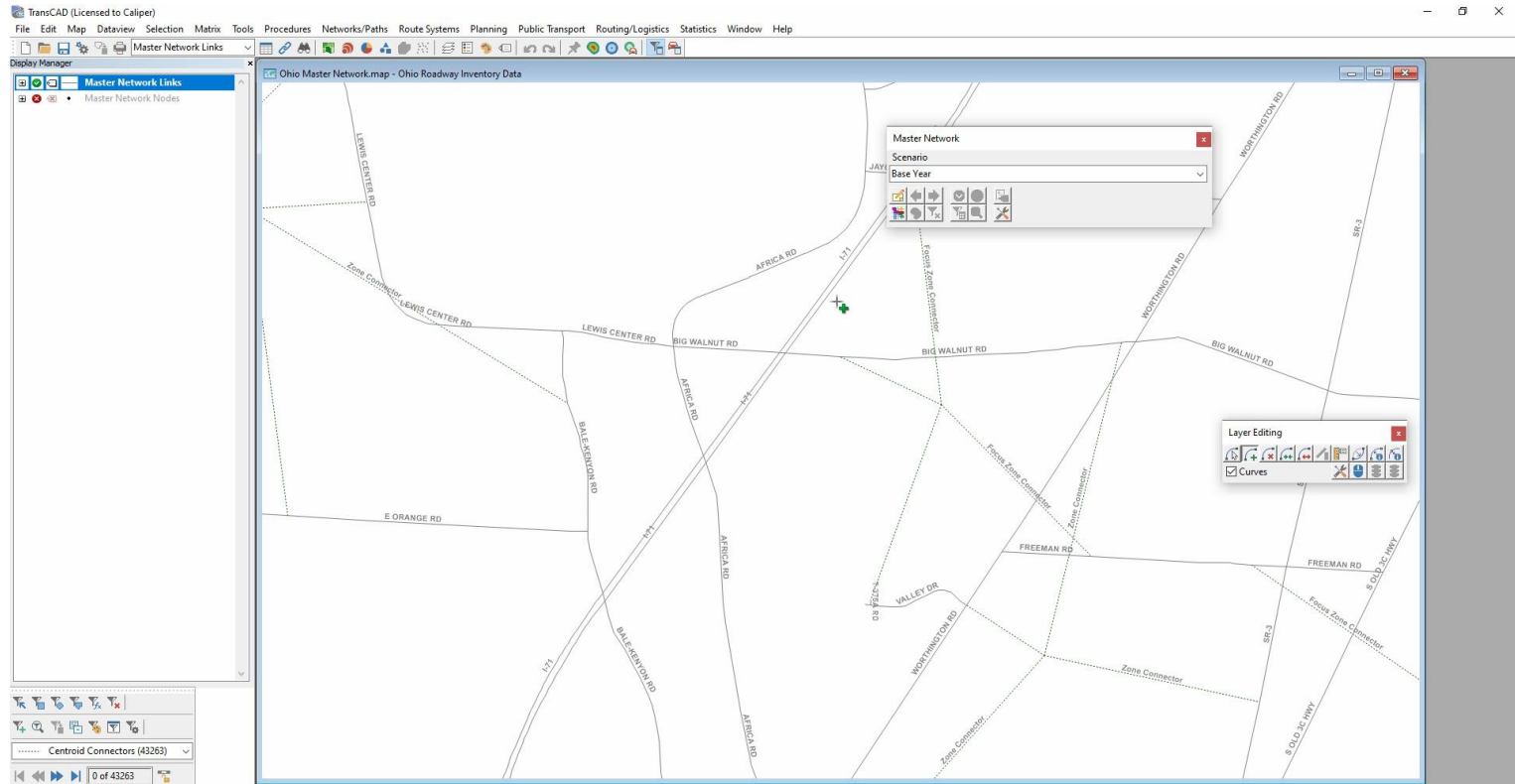
- Code projects once, use in any scenario
- Record and maintain evolution of road networks
- Manage geographic and attribute changes in a single line layer
 - Easy toggling between scenario only and all scenario features visible
- Maintain temporal as well as geographic consistency



TRANSCAD MASTER NETWORK FUNCTIONALITY



MASTER NETWORK EXAMPLE



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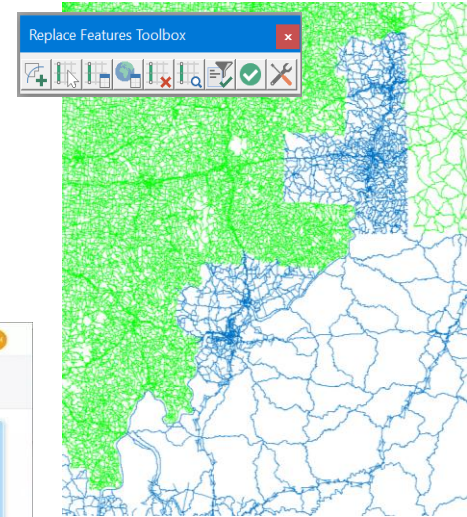
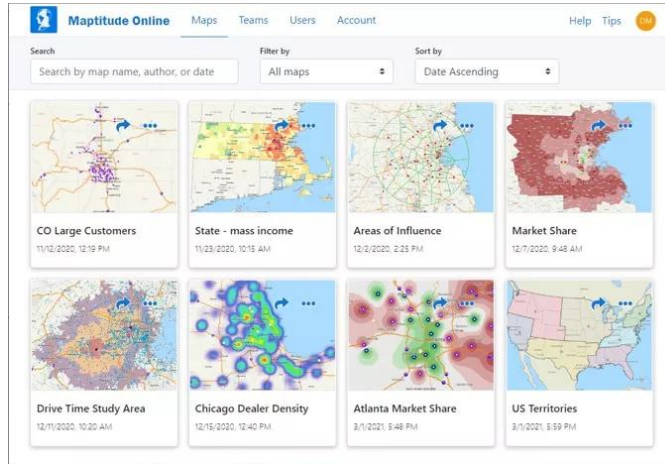
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A COLLABORATIVE ENVIRONMENT

- Check-out and check-in of subarea networks
 - Counties or custom subareas (e.g., project corridors)
- Web publishing and commenting applications
 - Incorporate web maps of scenario networks in plan documents, etc.
- Flexible hosting
 - your server or ours
 - your network or the web



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A UNIFIED SOLUTION IS POSSIBLE NOW

- Caliper can efficiently complete and improve Ohio's ARNOLD
 - Ensure routability
 - Improve geographic accuracy
 - Complete all roads coverage
- Easy management of multiresolution networks and zones
 - Custom tools for Ohio data schemes with limited scripting
- Easy management of projects and scenarios
 - Built-in functionality can handle network splits, additions, etc.
- Flexible tools for collaboration
 - Readily configurable to your desired architecture



CONTACTS

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